A Project Report on

"ResiPlan AI: An Intelligent System for Residential Floor Planning using AI/ML"

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Certificate

This is to Certify that the project report entitles

"ResiPlan AI: An Intelligent System for Residential Floor Planning using AI/ML"

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are bonafide students of this institute and the work has been carried out by them under the guidance of Prof. Ashutosh Kale and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University for the award of the degree of Bachelor of Engineering (Computer Engineering).

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Abstract

Designing residential floor plans is traditionally time-consuming, costly, and inaccessible for non-technical users, as existing tools like AutoCAD and Revit demand advanced skills, high processing power, and manual effort . This process, which depends on expert architects, is often complex and requires multiple revisions. Hence, there is a clear need for a simple, automated, web-based solution that can generate optimized floor plans with minimal user input

To overcome these challenges, our project, ResiPlan AI, proposes a web-based intelligent system using AI/ML to fully automate residential floor planning. The system, built on the MERN stack , allows users to provide simple inputs such as plot size, number of rooms, and preferred style via an interactive UI. These selections are converted into a text prompt for a Stable Diffusion AI model , which instantly generates optimized 3D floor plans. This approach makes residential planning fast, affordable, and accessible , allowing users to easily view, download, or regenerate designs.

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Chapter 1

Introduction

The acquisition or construction of a new home represents one of the most significant financial and personal investments an individual or family will make in their lifetime. At the heart of this endeavor lies the floor plan—the fundamental blueprint that dictates the home's functionality, flow, and character. A well-designed floor plan can enhance quality of life, optimize space, and reflect the unique needs of its inhabitants. Conversely, a poor layout can lead to daily inconveniences, inefficient use of space, and long-term dissatisfaction.

Traditionally, the process of creating this foundational plan is entirely walled off from the end-user. It is viewed as a purely technical task reserved for qualified architects and draftsmen. While professional expertise is invaluable for ensuring structural integrity and code compliance, this model creates a significant disconnect. Homeowners are often relegated to the role of passive observers, struggling to articulate their vision or visualize complex 2D drawings, only to discover misalignments between their expectations and the final product when changes are costly or impossible.

The desire for greater personal involvement in home design is growing. Individuals want the ability to experiment with ideas, compare different layouts, and understand the trade-offs of their choices before engaging professional services. This desire for creative agency and visualization has created a clear and unmet demand in the market: a tool that bridges the gap between a simple idea and a viable architectural concept.

While some consumer-grade design applications exist, they largely fail to solve

the core problem. These tools tend to be "digital sketchpads" or simple room builders. They may simplify the drawing process, but they still demand significant design knowledge from the user. They do not offer intelligent guidance, optimization, or automated generation. A user must still know the principles of good design: standard room dimensions, efficient traffic flow, structural feasibility, and logical adjacencies (e.g., placing a dining room near a kitchen). In short, they make it easier to draw a bad plan, but they do not help the user create a good one.

To solve this, our project, ResiPlan AI, introduces a web-based intelligent system that leverages the power of Artificial Intelligence (AI) and Machine Learning (ML) to automatically generate residential floor plans. ResiPlan AI is designed from the ground up to reverse the traditional model. Instead of requiring technical skill from the user, it asks for simple, intuitive inputs. Users provide basic parameters such as: Overall plot dimensions (length and width). The desired number and type of rooms (e.g., "3 bedrooms, 2 bathrooms, 1 kitchen"). Preferred styles or key relationships (e.g., "open-plan living," "master bedroom isolated"). From this simple set of constraints, the system's AI/ML core instantly produces an optimized 2D layout. The underlying models, trained on vast datasets of successful architectural plans, understand principles of spatial efficiency, lighting, and flow. The system not only draws the plan but designs it.

Furthermore, ResiPlan AI will provide an immediate 3D visualization, allowing the user to virtually walk through their new home. This approach makes residential planning easy, fast, affordable, and, for the first time, accessible to everyone, even for absolute beginners.

Chapter 2

Literature Survey

To build ResiPlan AI, we reviewed existing research on automated floor plan analysis and generation. The field has evolved from foundational datasets to highly advanced, text-driven generative models.

2.1 CubiCasa5K: A Dataset and an Improved MultiTask Model for Floorplan Image Analysis (2019)

This paper addresses the lack of large, representative datasets for floor plan research. It introduces CubiCasa5K, a large-scale dataset of 5,000 floorplan images. The samples are richly annotated with polygons for over 80 object categories. The availability of such large datasets is critical for training the deep learning models used in modern floor plan generation

The advantage of this system lies in its real-time scanning ability, user-friendly mobile interface, and efficient nutrient tracking features. However, its accuracy depends largely on string matching techniques, leading to occasional misclassification of complex ingredient names. Moreover, the system was limited to printed food labels and lacked the capability to analyze cooked or unpackaged food items.[1]

2.2 Graph-Constrained Vector Generation(2025)

This work presents House Diffusion, a novel approach that generates floor plans as vector graphics by denoising the 2D coordinates of room and door corners. This

method is significant because it outputs structured vector data, not just a raster image. However, its input relies on a "bubble diagram," a graph that defines room adjacencies. While effective, this graph-based input is still too technical for non-expert users.[2]

2.3 Text-to-Image Generation with Diffusion Models(2023)

Text semantics to controllable design: A residential layout generation method based on stable diffusion model" This research is highly relevant as it proposes a method for generating residential layouts using a Stable Diffusion model controlled by natural language text . It specifically incorporates ControlNet, which enables the generation of controllable layouts that adhere to design constraints. This paper validates the idea of using text-based prompts to create controllable and diverse designs, making the process accessible to non-professional[3]

2.4 ChatHouse Diffusion:Prompt-Guided Generation and Editing of Floor Plans(2023)

This paper introduces ChatHouse Diffusion, a system that leverages LLMs to interpret natural language input for both generating and, crucially, editing floor plans. Existing methods often lack flexibility, requiring a complete redesign if a user is dissatisfied. This approach allows users to make iterative adjustments with simple text commands (e.g., "add a kitchen" or "make this room bigger"), significantly enhancing design efficiency and user-friendliness.[4]

2.5 Generating floorplans for various building functionalities via latent diffusion model (2024)

This paper expands the text-to-image concept, demonstrating that latent diffusion models can generate floor plans for a wide variety of building types, including apartments, offices, and even stadiums. The model uses two inputs: a building footprint image and a text-based design brief. This confirms the flexibility and scale-agnostic capabilities of

diffusion models in architecture.[5]

2.6 Summary

The primary research gap is not in the models themselves, but in their accessibility. Most research focuses on the model architecture, leaving a need for a simple, end-to-end web application that makes this powerful technology available to non-technical users. Our project, ResiPlan AI, aims to fill this gap by translating simple user-interface selections (e.g., "3 bedrooms, 2 bathrooms") into the complex text prompts these advanced models require, hiding all technical complexity from the end-user

Chapter 3

Problem Statment

Designing residential floor plans is traditionally time-consuming, costly, and inaccessible for non-technical users, as it depends on expert architects using complex tools like AutoCAD and Revit. These existing tools demand advanced skills, high processing power, and significant manual effort. Consequently, people with little or no design knowledge face significant challenges in creating or visualizing their own house plans. Therefore, there is a clear need for a simple, automated, and web-based solution that can generate optimized residential floor plans with minimal user input

3.1 Objectives and Scope

Objectives

- To develop a fully automated residential floor planning system using AI/ML.
- To provide web-based access, reducing the need for heavy installations or high processing power.
- To generate optimized 2D floor plans and 3D visualizations instantly.
- To ensure efficient space utilization with considerations for energy efficiency (ventilation, lighting, and circulation).

Scope

- User-Friendly System: A web-based platform that even non-technical users can operate with zero design knowledge.

- Automated Floor Planning: AI/ML algorithms will generate 2D residential layouts automatically based on user inputs (plot size, rooms, style, etc.).
- Visualization Support: Provide 3D visualizations or simple renders of the generated layouts for better understanding.
- Customizable Inputs: Users can choose the number of bedrooms, bathrooms, kitchen type, living space, etc.

3.2 Summary

In this chapter, we introduced the problem of traditional floor planning, which is time-consuming, costly, and inaccessible for non-technical users . We discussed the necessity for an AI-powered solution to ensure an automated, fast, and affordable design process. The objectives and scope define how ResiPlan AI will use AI/ML and diffusion models to create an intelligent system that helps users generate their own optimized 2D and 3D floor plans.

Chapter 4

Analysis

This chapter describes the project plan adopted and determines the requirement analysis. We have implemented the project on the basis of Rapid Application Development (RAD) model and Model View Controller (MVC) model. The stake holders who participated in the requirement analysis process were the developers of Cognifront who will be among the end users of the M-Learning Framework for building M-Learning Applications.

4.1 Project Plan

4.1.1 Project Plan for semester I

To complete the research, system design, and initial proof-of-concept for the ResiPlan AI web application. This includes finalizing the architecture, modeling the system with UML, and developing a UI prototype with a functional AI generation pipeline.

Phase	Activity	Start Date	End Date	Group Members
1	Selection of Project Topic	15-07-2025	18-07-2025	Team
1	Problem Statement Finalization	26-08-2025	30-08-2025	Swapnil
1	Defined Project Objectives	26-08-2025	30-08-2025	Jayesh, Swapnil
1	Paper Presentation and Literature Survey	12-09-2025	15-09-2025	Team
1	Functional Requirement Specification (FRS)	20-07-2025	28-07-2025	Jayesh, Onkar
1	Design Prototype (UI and System Flow)	30-07-2025	10-08-2025	Team
1	Set Theory and Mathematical Model	12-08-2025	20-08-2025	Vedant, Onkar
1	UML Diagram Prototype	15-08-2025	25-08-2025	Team
1	UML Diagram	01-09-2025	10-09-2025	Team
1	Software Requirement Specification (SRS)	16-09-2025	25-09-2025	Team

Table 4.1: Project Plan for Semester I

4.1.2 Project Plan for semester II

The following Table 4.2 describes the project plan for semester II. It describes the various activities and accountability of the developers for the respective modules. Following are the major activities carried out in this plan :

- Define Programming Standards.
- Development of project in 3 Milestones.
- Formal Technical Review and Testing.

Phase	Activity	Start Date	End Date	Group Members
2	Defining Programming Standards	10-12-2011	15-12-2011	Team
2	Development of Milestone No.1	16-12-2011	05-01-2012	Team
2	Development of Milestone No.2	7-01-2012	02-02-2012	Team
2	Development of Milestone No.3	05-02-2012	29-02-2012	Team
2	Formal Technical Review	02-03-2012	10-03-2012	Team
2	Testing and Bug Fixing	22-03-2012	10-04-2012	Team

Table 4.2: Planner and Progress Report II for Maggie

4.2 Requirement Analysis

4.2.1 Necessary Functions

- Web App
- Prompt Generation
- AI Generation
- Display and Export
- User-Friendly Interface

4.2.2 Desirable Functions

- Allow users to "tweak" or modify the generated designs.
- Suggest eco-friendly or energy-efficient design options.
- Multilingual Support.

4.3 Summary

This project is a user-friendly MERN web app that automatically generates 3D residential floor plans using AI . It takes simple user selections (like room count) and uses the Stable Diffusion API to create a high-quality design . The system provides a fast and accessible solution for residential planning, allowing users to view, download, or export the generated layouts

Chapter 5

Design

This chapter describes the Software Requirement Specification (SRS) for the ResiPlan AI system. It explains the system architecture, external interface requirements, and the design constraints that guided the development. Additionally, it outlines the risk assessment strategy and includes data flow details that illustrate how data moves throughout the system components

5.1 Software Requirement Specifications

The Software Requirement Specification (SRS) defines the scope of the project, operating environment, user characteristics, design limitations, and overall system architecture of the ResiPlan AI application. It serves as a foundation for the software design and implementation process.

5.1.1 Project Scope

ResiPlan AI leverages generative AI and predictive modeling techniques to create floor plans for various residential buildings such as apartments, villas, and duplex houses. It allows users to input preferences such as plot size, number of rooms, and design style, and outputs optimized floor layouts. The system also supports visualization, space utilization analytics, and modification tools.

- Intelligent generation of 2D and 3D residential layouts
- Integration of ML models for pattern-based design prediction

- User-friendly interface for editing and exporting plans.
- Compliance with local building codes and design standards

5.1.2 Operating Environment

ResiPlan AI is designed to operate seamlessly on web-based platforms using a React frontend and Python (Flask) backend for efficient performance. It integrates diffusion-based AI models and uses MySQL databases to ensure scalable and reliable data management. The system requires a minimum of 4GB RAM and a PC with internet access, providing flexibility, portability, and smooth execution.

5.1.3 User Classes and Characteristics

This class represents the main consumers of the application, which are further broken down by their motivations

- Architects/Designers: Use the AI tools to speed up the floor plan creation process.
- Builders/Developers: Use the system to generate quick layouts for residential projects.
- Homeowners: Use the system to visualize personalized house plans based on their preferences.

5.1.4 Design and Implementation Constraints

- AI Model Dependency: The accuracy of the generated floor plans is entirely dependent on the trained ML models.
- Internet Dependency: The system requires a stable internet connection to execute the AI model and communicate with cloud-based APIs.
- Hardware Limitations: The rendering of large-scale plans is limited by the system's hardware.
- User Knowledge: The system assumes users have a basic understanding of residential layout concepts.

5.2 System Architecture

The user selects parameters on the React Frontend and clicks "Generate". This sends a JSON request to the Node.js Backend, which validates the input, creates an AI prompt, and calls the Hugging Face AI Model Layer. The Stable Diffusion 1.5 model (with optional ControlNet) generates the 3D floor plan image. This image is returned to the backend, then sent to the frontend for the user to view, export, or download, as shown in Fig 1.

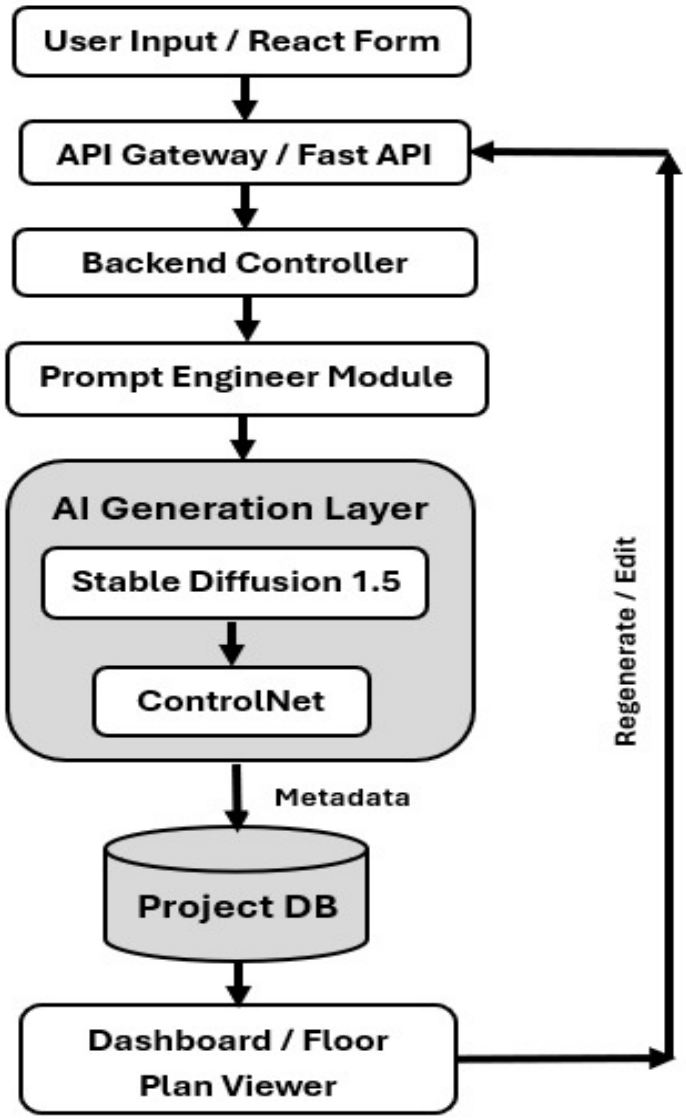


Figure 5.1: System Architecture

5.2.1 Frontend (React)

This is the client-side, user-facing part of the application, built using React.js. This component is responsible for all user interactions. It features an interactive "Parameter Selection UI" where the user inputs their requirements, such as the number of bedrooms, bathrooms, kitchen, and preferred architectural style. Once the plan is generated, the frontend displays it in the "Floor Plan Viewer". This module is designed to be highly user-friendly, requiring no prior design knowledge.

5.2.2 Backend (Node.js / Express API)

The Backend serves as the central processing unit for the application, built as a RESTful API using Node.js/Express. Its primary role is to receive the simple JSON request from the React frontend. It then uses an "Input Validator" to ensure the data is correct. Its most critical component is the "Prompt Generator", which translates the user's simple selections (e.g., "3 bedrooms," "modern style") into a rich, descriptive text prompt that the AI model can understand and execute.

5.2.3 AI Model Layer (Hugging Face)

The AI Model Layer is an external API service, specifically Hugging Face. This layer acts as a crucial intermediary, decoupling our lightweight backend from the heavy computational load of the AI model. The backend sends its generated text prompt to this API. The Hugging Face layer handles the request, validates it, forwards it to the generative model, and, after processing, returns the final 3D floor plan image to our backend.

5.2.4 AI Model (Stable Diffusion 1.5)

The core "brain" of the system is the Stable Diffusion 1.5 generative model. This is a pre-trained deep learning model that excels at creating high-quality, photorealistic images from text descriptions. In our system, it takes the detailed text prompt (e.g., "A 3D photorealistic render of a modern floor plan with 3 bedrooms...") and generates a complete 3D floor plan image that matches those specifications.

5.2.5 Data Store

The Data Store is the system's database, specified as MySQL or Firestore. This component is responsible for persisting data. As shown in the architecture, its primary functions are to store user history, logs, and user preferences. This allows users to save their generated plans, view past designs, and maintain their account information

5.3 External Interface Requirement

5.3.1 User Interfaces

- Interactive dashboard for plan generation.
- Drag-and-drop editor for customization
- Export button for PDF/DWG outputs.

5.3.2 Hardware Interfaces

- Requires a device with at least 4 GB RAM and a modern web browser

5.3.3 Communication Interfaces

HTTPS protocol for secure communication HTTPS and RESTful APIs to access databases such as FSSAI or FDA and RESTful APIs for AI model requests and responses.

5.4 Software System Attribute

- **Reliability:** System uptime of 99 percent and Automatic recovery from minor failures.
- **Availability:** The system is specified to have a system uptime of 99 percent.
- **Portability:** Cross-platform support (Windows, macOS, Linux)

5.5 Functional Requirement

- **User Input:** System must capture parameters like rooms, size, and style via a web UI.
- **Generation:** System must use a backend to convert user inputs into a prompt, call the Hugging Face (Stable Diffusion) API, and generate a 3D floor plan
- **Display:** System must display the generated 3D plan in the UI
- **Export:** User must be able to download/export the plan in PNG formats.
- **Iteration:** User must be able to regenerate or modify the plan.
- **Storage:** System must save user history, preferences, and generated plans

5.6 Non-Functional Requirement

- **Performance:** Floor plan generation within 10 seconds for standard inputs and Support up to 500 concurrent users.
- **Security:** Encrypted user data storage and transmission (SSL/TLS).
- **Reliability:** Automatic recovery from minor failures.
- **Usability:** Intuitive and responsive UI/UX

5.7 Summary

In this chapter, we described the Software Requirement Specifications (SRS), detailing the scope of the ResiPlan AI project, the operating environment, and the user classes and characteristics involved in the system. We also explained the system architecture, external interface requirements.

Chapter 6

Modeling

This chapter presents the various modeling techniques used to represent the behavior, structure, and interactions within the ResiPlain AI system. It focuses on understanding how different users interact with the system and how the system responds through specific functionalities. Modeling helps in visualizing, specifying, and documenting the design of the system to ensure a clear understanding of requirements and operations.

6.1 Use Case Diagram

A Use Case Diagram is a type of behavioral diagram defined by UML (Unified Modeling Language) that represents the interaction between users (actors) and the system. Its primary purpose is to provide a graphical overview of the functionalities offered by the ResiPlan AI system, depicting how different users achieve their goals through system features.

In the ResiPlan AI project, the Use Case Diagram illustrates how users, architects, and developers interact with the system to perform tasks such as entering design parameters, generating floor plans, visualizing layouts, modifying designs, and managing saved plans.

Modeling Elements in ResiPlan AI Use Case Diagram

- **User/Consumer:**The primary actor is the User

- **System:** The system is the ResiPlan AI System.
- **Core Process:** Select Room Parameters includes the Generate 3D Floor Plan use case. Generate 3D Floor Plan includes the View Generated Plan use case.

Optional Actions:

- Download/Export Design.
- Regenerate / Modify Layout

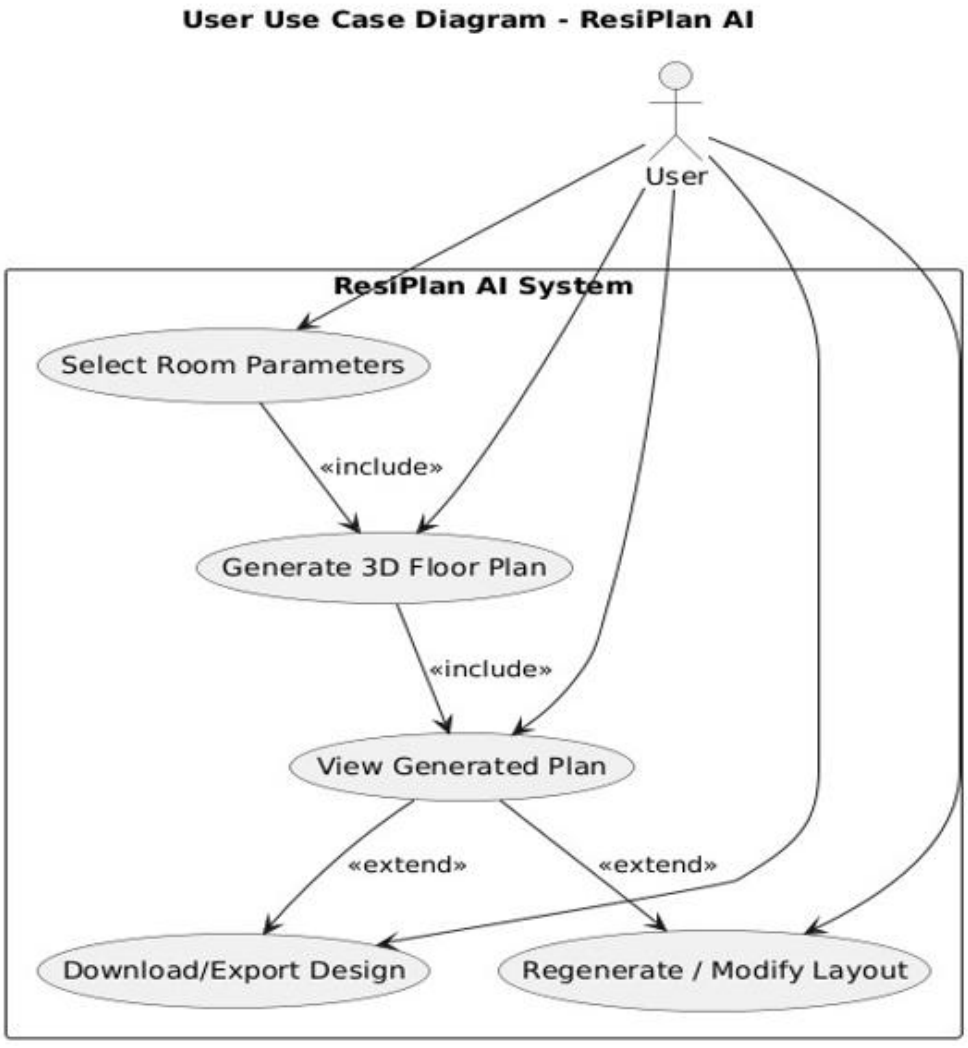


Figure 6.1: Use Case Diagram

6.2 Class Diagram

- **User:** Represents the person using the app. Manages user details and actions like viewing or modifying plans.
- **BackendController:** The main "brain" of the server. It handles user requests, sends tasks to the AI model, and manages database operations.
- **AIModel:** Represents the Stable Diffusion model. Its job is to process the input prompt and generate the final image.
- **Database:** Handles all data storage, such as saving, fetching, or updating floor plans.
- **Evaluator:** An optional class to score the quality of a generated plan based on metrics like accuracy and geometry.
- **FloorPlan:** The central data object that holds all information about the generated plan, including its ID, 2D/3D layouts, and room list.

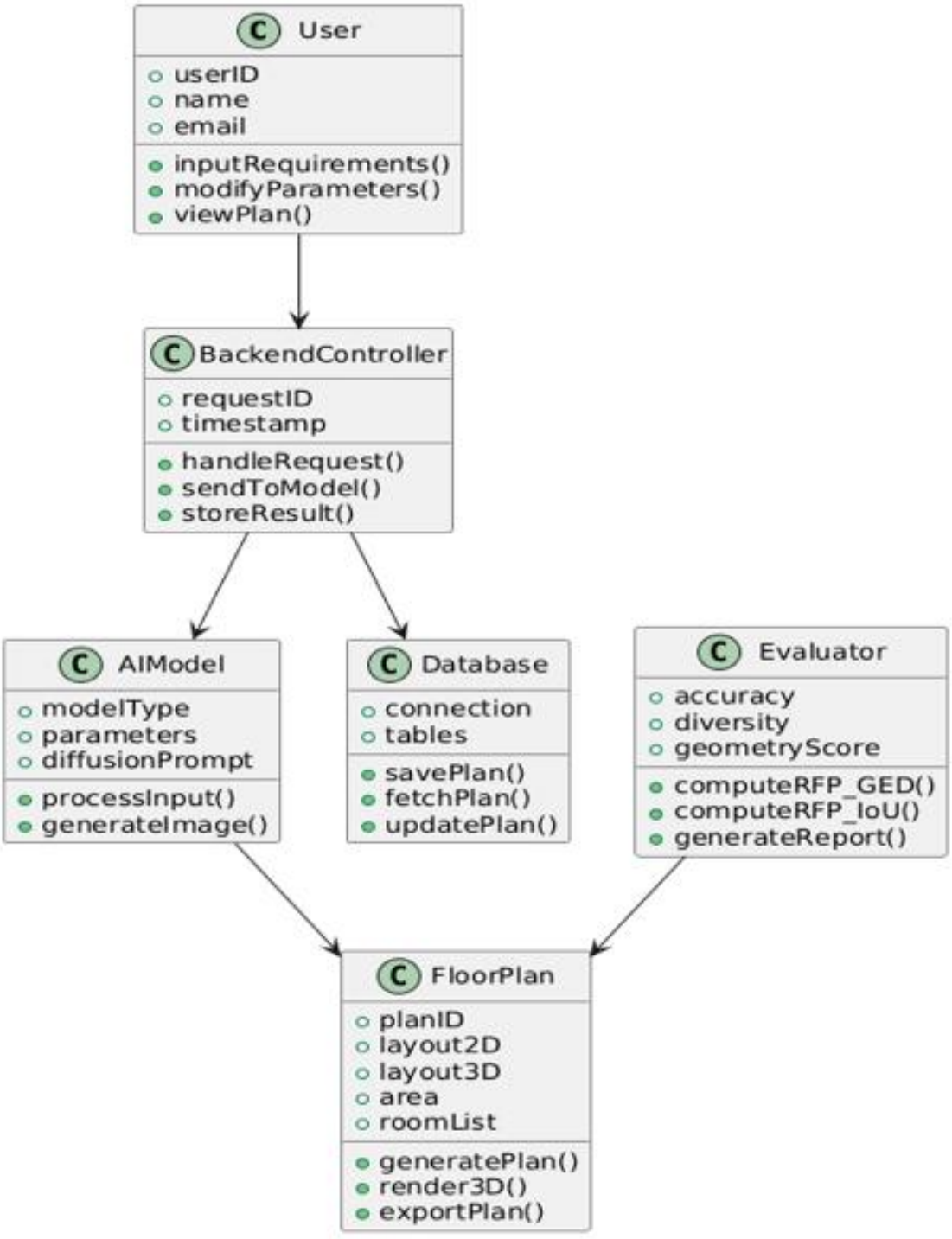


Figure 6.2: Class Diagram

6.3 Activity Diagram

The workflow begins when the User opens the web app. The user then proceeds to Enter plot size, rooms, and style and clicks Submit request.
The BackendController receives the input and then Converts the input to a diffusion prompt. This prompt is Sent to the AI Model, which generates the floor plan.

The system then Displays the 2D/3D plan to the user. At this point, a decision is made based on whether the "User is satisfied?":
If Yes: The user can Download or export the plan, which is then Stored in the database, and the process ends.
If No: The user can Modify inputs and then Regenerate the plan

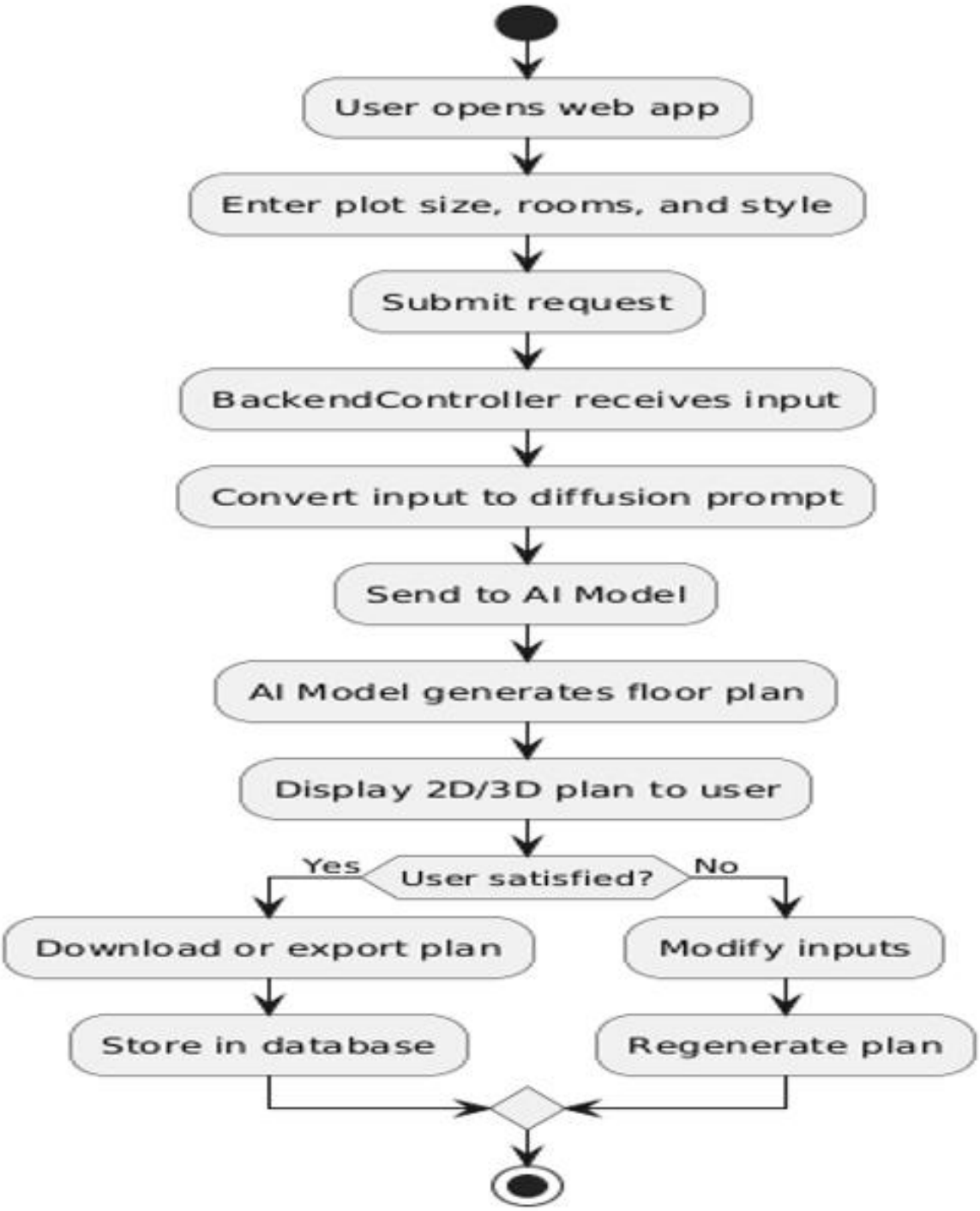


Figure 6.3: Activity Diagram

6.4 Sequence Diagram

This flow starts when the User inputs their design preferences (like plot size, rooms, and style) into the Web UI. The Web UI bundles this information and sends a Send design request message to the BackendController.

The BackendController receives this request, processes it, and generates a prompt. It then sends a Generate plan (prompt) message to the AI Model. The AI Model generates the floor plan image and sends it back as a Return generated layout message to the BackendController. The BackendController then forwards this plan to the Web UI with a Display 2D/3D plan message. Finally, the Web UI renders the plan, and the User can see the Show generated plan result.

After the plan is generated, the User initiates a Request to save or download via the Web UI. The Web UI sends a Save plan message to the BackendController.

The BackendController then sends a Store layout and metadata message to the Database. The Database saves the information and replies with a Confirmation message.

Upon receiving this confirmation, the BackendController sends a Notify success message back to the Web UI, which then informs the User that the Operation [was] successful.

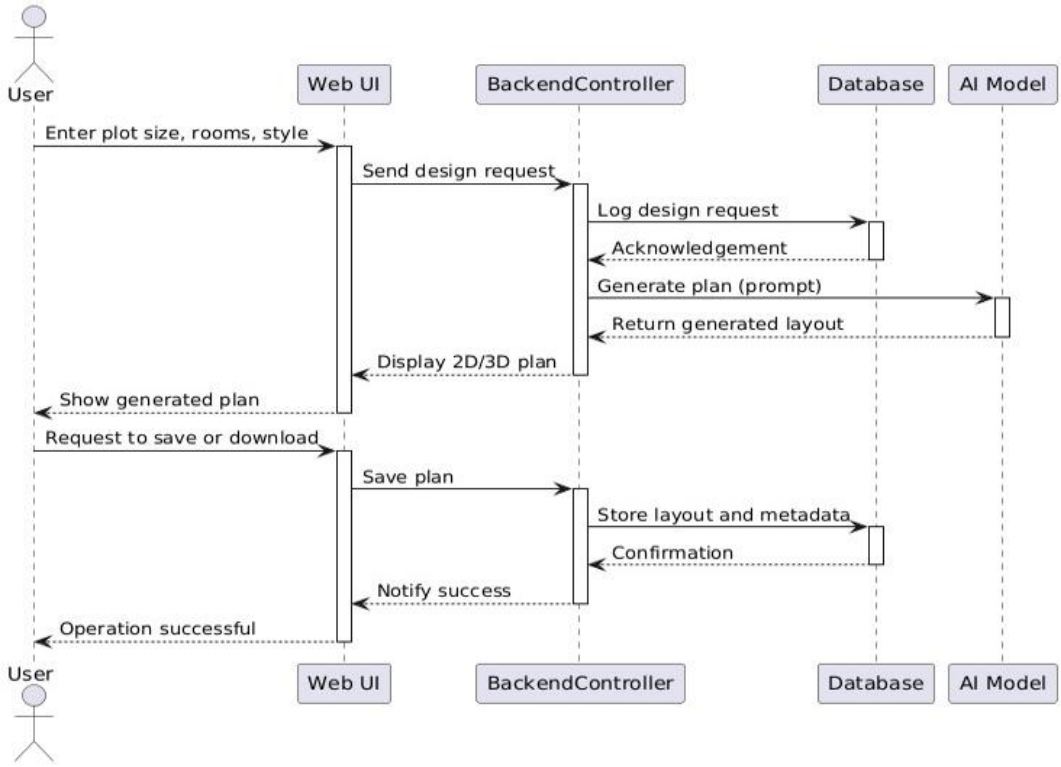


Figure 6.4: Sequence Diagram

6.5 Summary

Thus we saw the various modeling techniques used for the design of ResiPlan AI application.

Chapter 7

Implementation and Results

This chapter presents the implementation details and working of the ResiPlan AI system for automated residential floor planning using AI/ML. It also includes system outputs and snapshots demonstrating the generated floor plans and user interface.

7.1 Implementation Details

This section describes the various development phases of the ResiPlan AI system along with the implementation methods used in each phase.

Phase 1: Requirement Analysis

- Identified user requirements such as number of rooms, plot size, layout orientation, and design preferences.
- Defined system objectives including accuracy of generated plans, user-friendly interface, and scalability of the system.

Phase 2: Initial Prototype Development

- Developed a basic prototype using direct text-to-image generation.
- Generated floor plans using simple prompts without constraints.
- Observed issues such as:
  - Unstructured layouts

  - Incorrect room placement
  - Lack of proper labeling and alignment

Phase 3: Prompt Engineering Optimization

- Designed structured prompts with architectural constraints such as:
  - "Top view", "2D floor plan", "blueprint style"
  - Specific room definitions like bedroom, kitchen, and bathroom
- Improved the semantic understanding of the AI model for better output quality.

Phase 4: Backend Development

- Developed backend using Node.js and Express.js.
- Shifted API handling from frontend to backend for better security.
- Implemented:
  - Secure API key management
  - Error handling and logging
  - Retry mechanisms for failed requests
  - Standardized API responses

Phase 5: Frontend Development

- Developed an interactive user interface using React.js.
- Features included:
  - Input forms for room selection, size, and orientation
  - Option to select render style (2D / 3D)
  - Real-time display of generated floor plans

Phase 6: Model Integration

- Integrated Stable Diffusion model through API.
- Tested multiple model variants for performance.
- Selected optimized model for structured and accurate layout generation.

Phase 7: Image Post-Processing

- Applied image resizing and normalization techniques.
- Maintained consistent resolution and clarity of generated images.
- Improved visualization quality for frontend display.

Phase 8: System Integration

- Integrated frontend, backend, and AI model into a complete system.
- Implemented workflow:
  - User Input $\rightarrow$ Backend $\rightarrow$ AI Model $\rightarrow$ Output $\rightarrow$ Frontend
- Ensured smooth communication and data flow across all modules.

7.2 Results

The snapshots below are taken on the mobile device itself having Android 2.3.3 operating system, screen of 5 inches with a resolution of 800 x 480.

Following are the snapshots of M-Learning Application:

7.3 Summary

The implementation plan transformed the system from a basic prototype into a robust AI-powered application. Through systematic improvements in prompt engineering, backend architecture, and model optimization, the system achieved higher accuracy, reliability, and usability.

Chapter 8

Testing

This chapter includes the details of Formal Technical Review conducted for the ResiPlan AI system and describes the evaluation process carried out during development. It also includes the test plan adopted for validating the performance, accuracy, and reliability of the AI-based floor plan generation system.

8.1 Formal Technical Review

A formal technical review was conducted to evaluate the performance, reliability, and accuracy of the proposed AI-based floor plan generation system. The review focused on both functional correctness and AI output quality.

The system was evaluated on the following parameters:

- Structural correctness of generated layouts
- Semantic understanding of user inputs
- Visual clarity and readability
- API reliability and response handling
- System scalability and modularity

The review process included multiple iterations of testing with varied input configurations (room types, dimensions, and orientations). Results showed a significant improvement in output consistency and system robustness after implementing structured prompts and backend optimization.

8.2 Test Plan

A detailed test plan was designed to validate the system under different scenarios.

Test Scenarios

- Different house configurations: 1BHK, 2BHK, 3BHK
- Plot sizes: Small ($\leq$1000 sq ft), Medium (1000–2000 sq ft), Large ($>$2000 sq ft)
- Orientation: North, South, East, West
- Vastu compliance enabled/disabled
- Rendering styles: 2D Sketch, Photorealistic 3D

Testing Types

- Functional Testing: Verified correct generation of floor plans based on inputs
- Performance Testing: Measured response time and API success rate
- Usability Testing: Evaluated UI responsiveness and user interaction
- Stress Testing: Tested system under repeated and concurrent requests

8.2.1 Performance Comparison

Table 8.1: Performance Comparison

Metric	**Initial System**	**Improved System**
Layout Structure Accuracy	∼55%	∼80%
Room Label Accuracy	∼50%	∼75%
API Success Rate	∼70%	∼95%
Visual Clarity	Medium	High
Average Response Time	∼8–10 sec	∼4–6 sec

8.2.2 Evaluation Metrics

The following metrics were used for evaluation:

- **Structural Accuracy:** Measures correctness of room arrangement and layout flow.
- **Semantic Accuracy:** Evaluates how well the AI understands user input.
- **Visual Accuracy:** Assesses clarity and readability of generated images.
- **System Reliability:** Measures API success rate and error handling.
- **Response Time:** Time taken to generate output.

8.2.3 Test Result and Analysis

- Structured prompts significantly improved spatial alignment and room placement.
- Backend validation reduced incorrect and ambiguous inputs.
- Model optimization reduced noise and unwanted artifacts in generated images.
- Response time improved due to efficient API handling and reduced retries.

8.3 Summary

The testing phase confirms that the improved system is significantly more accurate, reliable, and efficient compared to the initial prototype. The integration of structured prompt engineering and backend optimization plays a crucial role in achieving these improvements.

Chapter 9

Technical Specifications

In this chapter we will discuss the advantages and limitations of the M-Learning Framework. We will also go through the applications of the framework and have a brief study about the technical requirements.

9.1 Advantages

The system provides several technical and practical advantages:

- Automated generation of floor plans using AI
- Significant reduction in manual design effort
- Faster generation compared to traditional methods
- Supports both 2D and 3D visualization
- Flexible input parameters (rooms, size, orientation)
- Improved accuracy through structured prompt engineering
- Scalable architecture using MERN stack
- Secure API handling via backend integration

9.2 Limitations

Despite improvements, the system has the following limitations:

- Lack of exact architectural precision and measurements
- Generated layouts may not comply with professional CAD standards
- Dependence on Stable Diffusion model outputs
- Limited customization for complex Floor designs
- Possible inconsistencies in highly detailed layouts

9.3 Applications

The proposed system can be applied in multiple domains:

- Residential Floor planning
- Real estate visualization
- Interior design prototyping
- Architectural education and training

9.3.1 Hardware Requirements

Component	**Requirement**
Processor	Intel i5 or higher
RAM	Minimum 8 GB
Storage	At least 10 GB free space
GPU (Optional)	NVIDIA GPU for faster processing

Table 9.1: Hardware Requirements

9.3.2 Software Requirements

Component	**Technology Used**
Frontend	React.js
Backend	Node.js, Express.js
Database	MongoDB
AI Model	Stable Diffusion
API Tools	Axios / Fetch
Development Tool	Visual Studio Code
Operating System	Windows / Linux / macOS

Table 9.2: Software Requirements

9.4 Summary

The technical specifications demonstrate that the system is robust, scalable, and suitable for real-world applications. The integration of AI with web technologies enables efficient and automated floor plan generation, although further improvements are required for architectural precision.

Chapter 10

Future Scope

- Integration of deep learning and generative design models (e.g., GANs, GNNs) for producing more realistic and optimized layouts.
- Support for multiple languages to expand usability.
- Voice-Based Interaction: Implement voice-based design commands to allow users to modify or create plans using natural language.
- Global Accessibility: Add support for multiple languages to expand the system's usability to a global audience.

Chapter 11

Conclusion

The ResiPlan AI system successfully demonstrates how Artificial Intelligence and Machine Learning can transform the traditional process of residential floor planning into an automated, efficient, and user-friendly solution. By leveraging diffusion-based models and a web-based architecture, the system enables users to generate optimized 2D and 3D floor plans using simple inputs such as plot size, room requirements, and design preferences.

The project effectively bridges the gap between non-technical users and complex architectural design tools, reducing dependency on expert knowledge and manual effort. Through iterative improvements in prompt engineering, backend integration, and user interface design, the system achieves reliable performance, faster generation, and improved usability.

Although certain limitations exist in terms of architectural precision and customization, the system provides a strong foundation for future enhancements. Overall, ResiPlan AI proves to be a scalable, accessible, and innovative solution for modern residential planning.

Appendix A

Glossary

- **Maggie:** We have code named our project M-Learning Framework for Multiple Platforms as Maggie.
- **M-Learning:** The application of small, portable, and wireless computing and communication devices like Laptops with Wireless LAN technology, mobile phones, Personal Digital Assistants (PDAs) with Bluetooth or IRDA interfaces for communication and computing.
- **TOC:** The term stands for Table of Contents which is a collection of contents. It can be used by the user to view the collection of learning materials available in the application.
- **OEM:** The term stands for Original Equipment Manufacturer, it is used to generate a logo for a particular application. Once an OEM logo is generated all the responsibilities are upto the manufacturer i.e. if any problem arises the manufacturer has to look after it till the problem is solved.

Appendix B

Achievements

- **First prize at CSI's National Level Discover Thinking**

  We had participated in **Computer Society of India's (CSI)** National Level Project Competition Discover Thinking sponsored by **Microsoft**. The contest had 3 rounds, 1st round at CSI chapter level held at particular colleges, 2nd round at CSI regional level held at Pune, and final round was at national level held at Coimbatore, Tamilnadu. Among 90 projects across the country our project was ranked at the first place after rigorous evaluation conducted by an eminent panel of judges from industry and educational background.

- **Published Research Paper in IJSER 2012**

  We have published a research paper titled **M-Learning Framework for Multiple Platforms** in **International Journal of Scientific and Engineering Research (IJSER)**. Our paper is published in the May edition of IJSER 2012.

Appendix C

Plagiarism Report

Appendix D

Team Information

Table D.1: Team Information

Sr.No.	Name of Member	Email ID	Mobile Number	Role in the Project
1				
2				
3				
4				

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